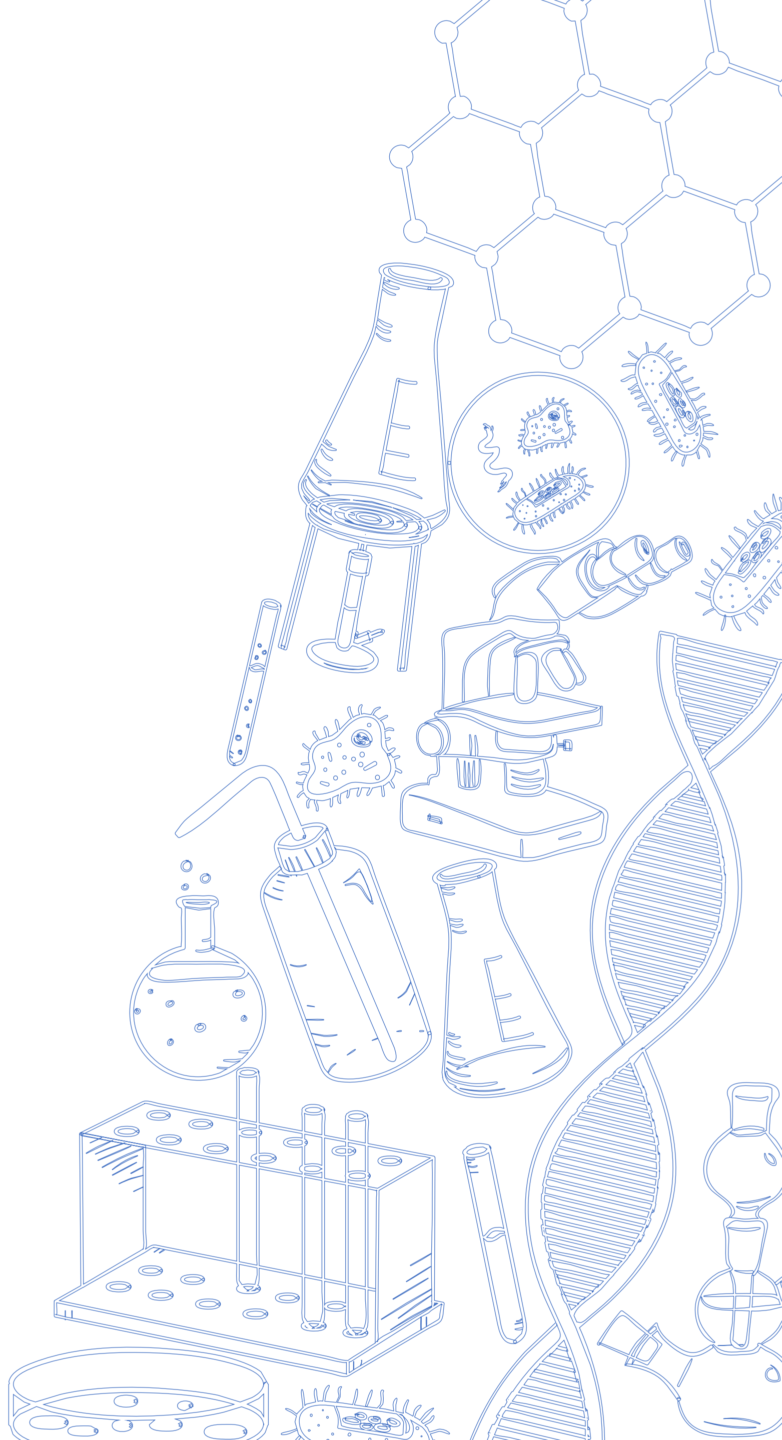
Science G10

Chemical Reactions
Class 1

Agenda

1. Course Outline
2. Classkick Portfolio Account Setup
3. Diagnostic Test
4. Lesson 1
   - Quick Review of G9 Chemistry
   - Lewis Dot Diagrams
   - Ionic Compounds & Nomenclature
   - Covalent Bonding & Nomenclature



Chemistry

- **Chemistry** is the study of matter and the changes it undergoes
- **Matter** is anything that has mass and volume

Physical & Chemical Properties

Physical Property can be measured and observed without changing the composition or identity of a substance

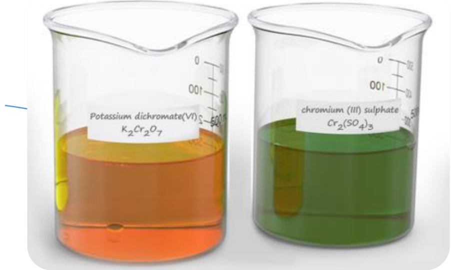
- Colour
- Density
- Solubility
- Melting point

Chemical Property can be observed when a substance changes into one or more new substances

- Flammability
- Bleaching
- Corrosion
- Rusting

Possible Evidence of Chemical Changes

New colour appears



New Solid

Solid material (precipitate) is formed

$A \neq B$

$A \longrightarrow B + \text{energy}$

Heat or light is produced or absorbed

Bubbles or gas form

$H_2O_{(l)}$ $H_2O_{(g)}$

Change is difficult to reverse

Checkpoint

Classify the following as a physical or chemical property:

L → G

a) Liquid nitrogen boils at -196°C Phys

b) Propane, leaking from a tank, ignites easily Chem

c) Silver jewelry tarnishes in air Chem

d) Spilled oil floats on the surface of water Phy

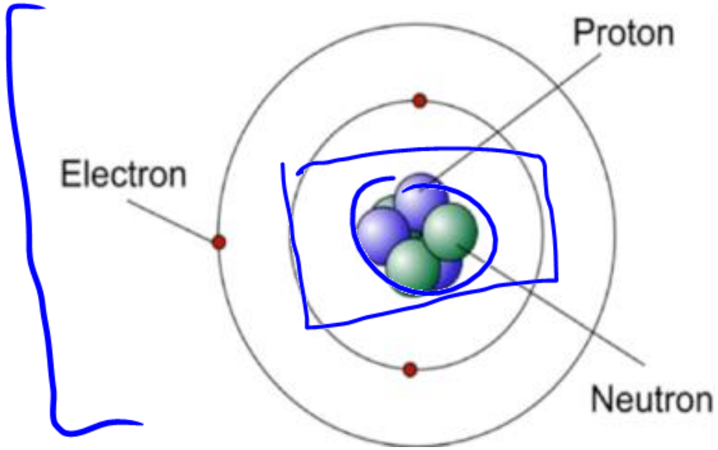
e) Meat darkens when it is heated on a grill Chem

f) Sulfur trioxide changes to sulfuric acid in the atmosphere

SO_3 H_2SO_4

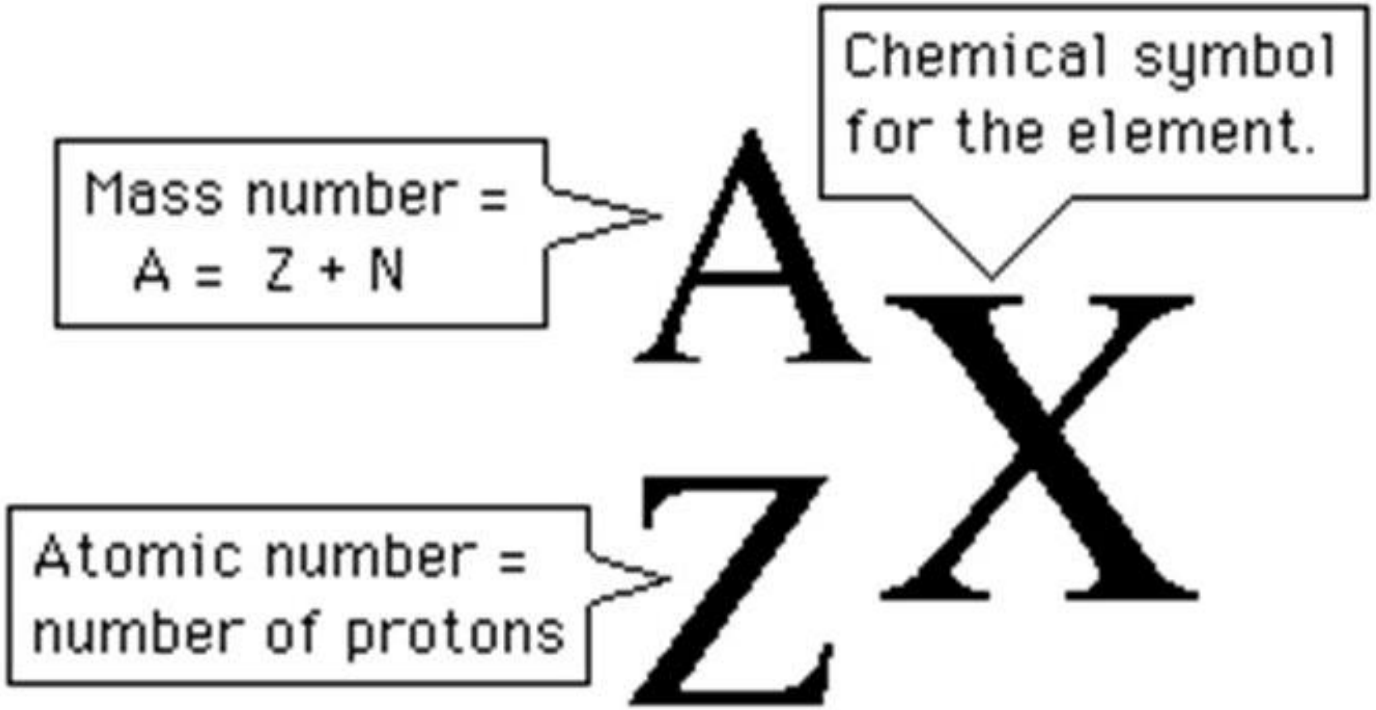
Atomic Structure

- An atom consists of:



	Proton	Neutron	Electron
Electrical Charge	+	0 no charge	-
Relative Mass	1	1	1/2000
Symbol	p^+	n^0	e^-
Location	Nucleus	Nucleus	Outside Nucleus

H



- **Mass Number (A)** is the total number of neutrons and protons present in the nucleus of an atom of an element
- **Atomic Number (Z)** is the number of protons in the nucleus of each atom of an element

Terms to Know

$10 + (-9)$
$= +1$

- **Isotopes** are atoms with the same atomic number but different mass numbers
- **Ions** are atoms with the same atomic number but a different number of electrons
  - **Cation** is a positively-charged ion
  - **Anion** is a negatively-charged ion
  - Note: Atoms become ions when there is a loss/gain of **electrons** NOT protons

3 3
p + n

3 4
(p + n)
mass #

Isotopes

	Li-6	Li-7
Protons	3	3
Electrons	3	3
Neutrons	3	4

Ion

	Na	Na^+
Protons	11	11
Electrons	11	10
Neutrons	12	12

+1
1 less electron

13

Checkpoint

$9.01 \xrightarrow{\text{rounding}} 9$

$12.5 \rightarrow 12$

$13.5 \rightarrow 14$

List the number of p^+, n^0, and e^- for:

$+9$

$x + y = -1$

$y = -1 - 9$

$y = -10$

a) ${}^{9}_{4}$Beryllium   b) Phosphorus   c) Fluoride ion (F^-)

a)

$(p + n) = 9$

$p = 4$

$n = 9 - 4$

$= 5$

$4\ p^+$

$5\ n^0$

$4\ e^-$

b)

$15\ p^+$

$16\ n^0$

$15\ e^-$

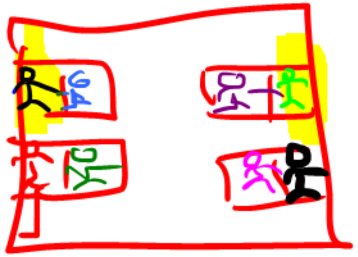
c)

$9\ p^+$

$10\ n^0$

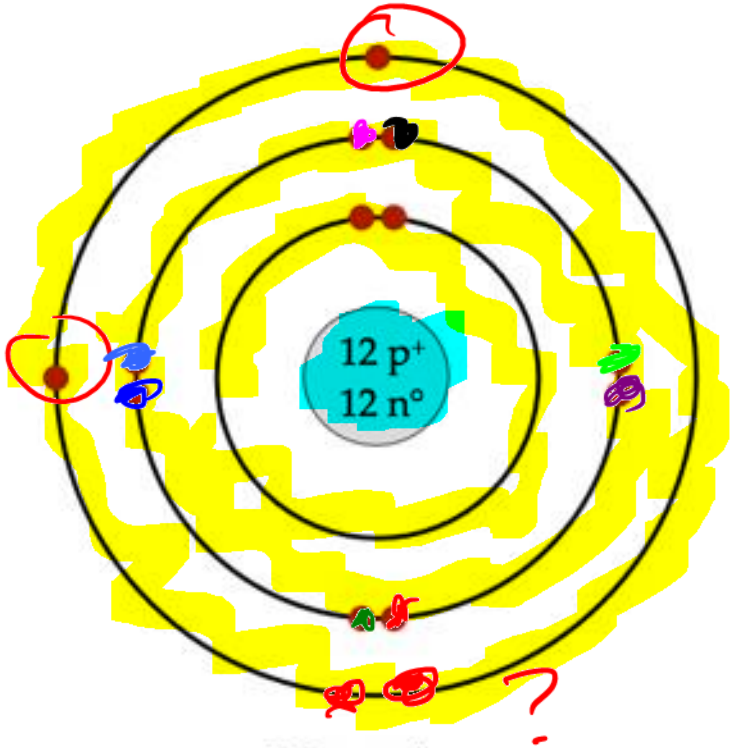
$10\ e^-$

Bohr-Rutherford Diagrams



door bus

1. Determine the number of protons, neutrons, and electrons
2. Draw the nucleus and write the number of protons and neutrons inside
3. Add the electrons to the outer shells:
    - 1st shell = 2 electrons
    - 2nd shell = 8 electrons
    - 3rd shell = $\boxed{8}$ electrons   4 pairs
    - 4th shell = 18 electrons



Magnesium

single first then
pair

STOP Checkpoint STOP

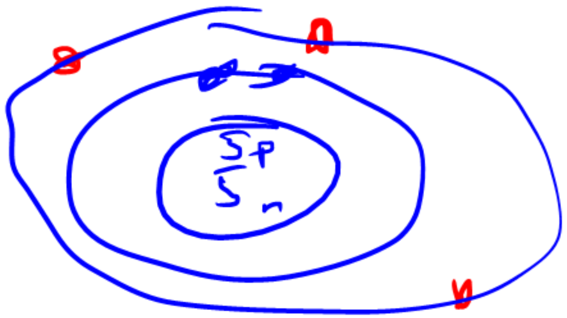
Draw the Bohr-Rutherford Diagram for:

a) Boron-$\boxed{10}$

$5e^- - 2 = \underline{3}$

$10 = p + n$ $\quad n = 10 - 5 = 5$

$5 = p$

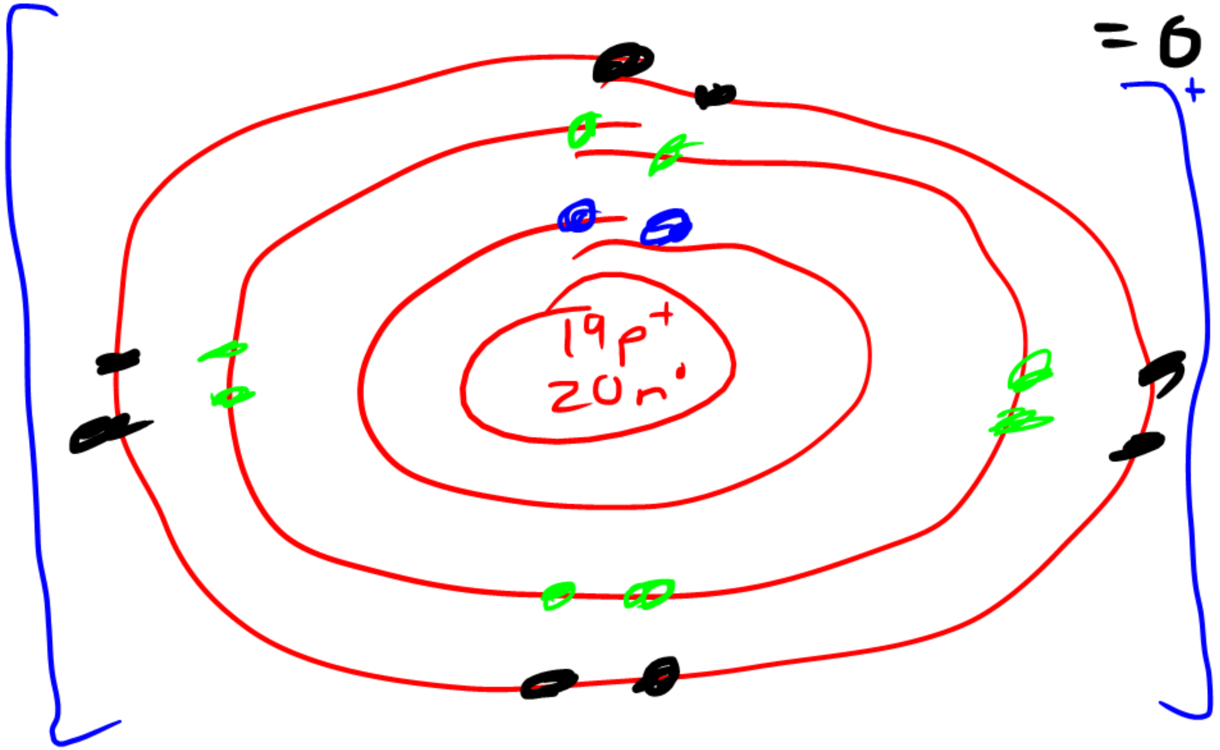


b) K^+

$18e^- - 2 = 16 - 8 = 8$

-8

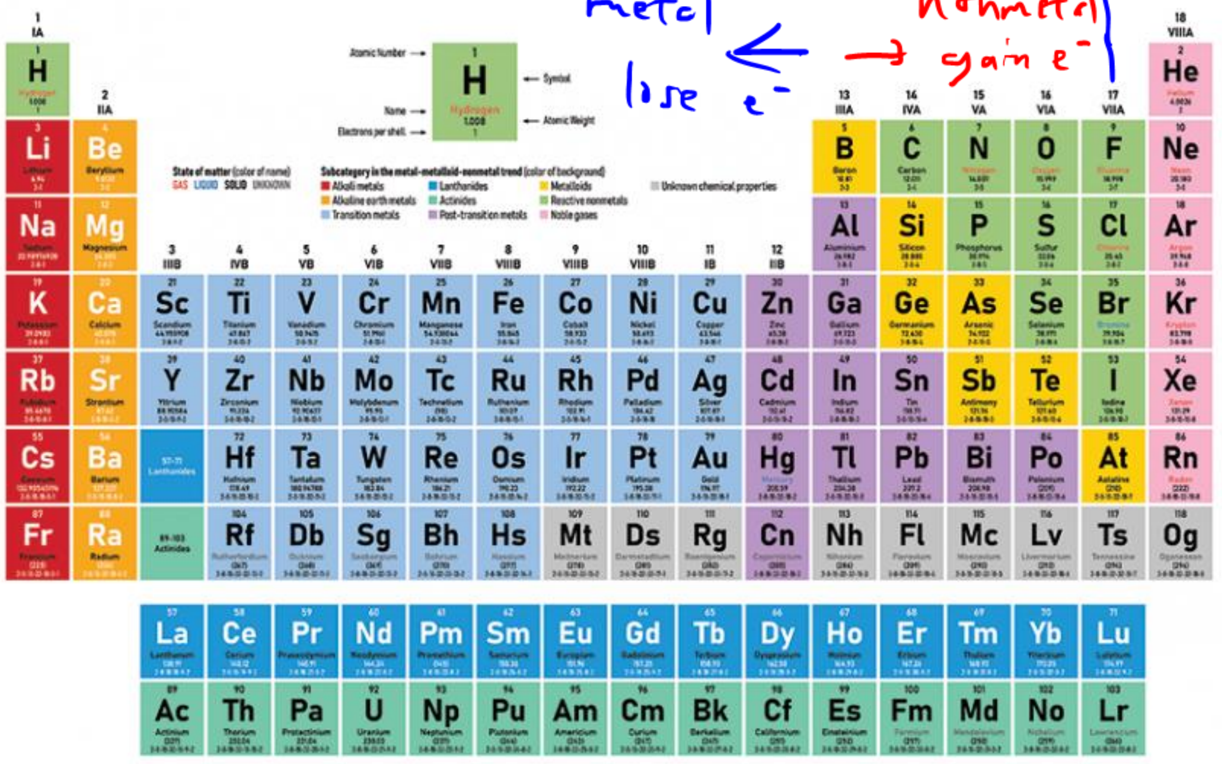
$= 0$



Patterns in the Periodic Table

- **Group/Family** is a column in the periodic table
- **Period** is a row in the periodic table

metal
lose e^-
nonmetal
gain e^-
halogens



	Alkali metals
	Alkaline earth metals
	Transition metals
	Post-transition metals
	Metalloids
	Nonmetals
	Noble gases
	Lanthanides
	Actinides
	Unknown chemical properties

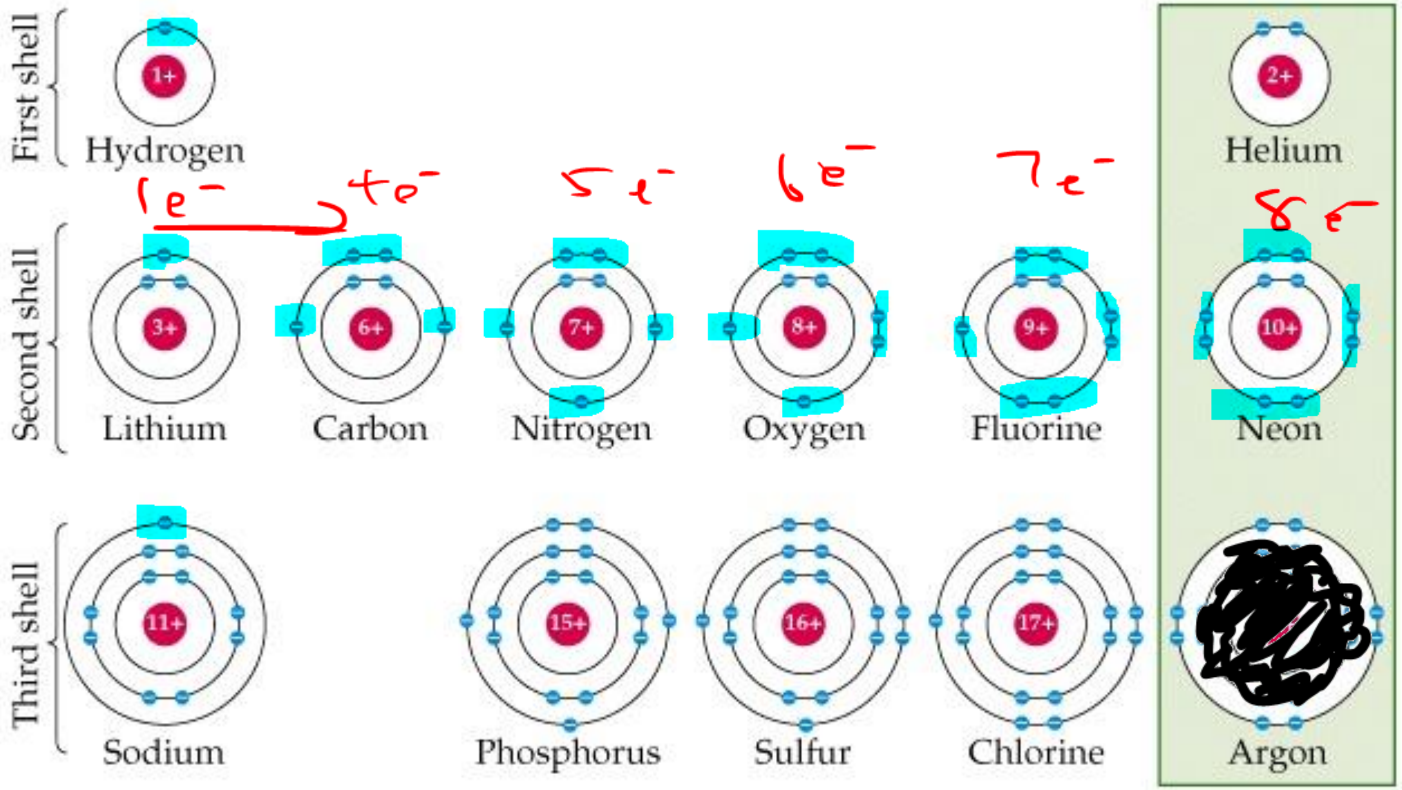
Trends in the Periodic Table

- Atomic number steadily increases across a period
- Number of energy levels (shells) steadily increases down a group
- Metals tend to lose electrons
- Nonmetals tend to attract electrons
- Number of valence electrons (outermost electrons) are the same within a group and increase steadily across a period

Reactivity

Noble gases are stable due to a complete valence shell

- Valence electrons determine the reactivity of an element and how compounds are formed
- Elements tend to lose or gain electrons valence electrons to form bonds and achieve stability



Alkali metals are very reactive; needs to lose $1e^-$

Halogens are very reactive; needs to gain $1e^-$

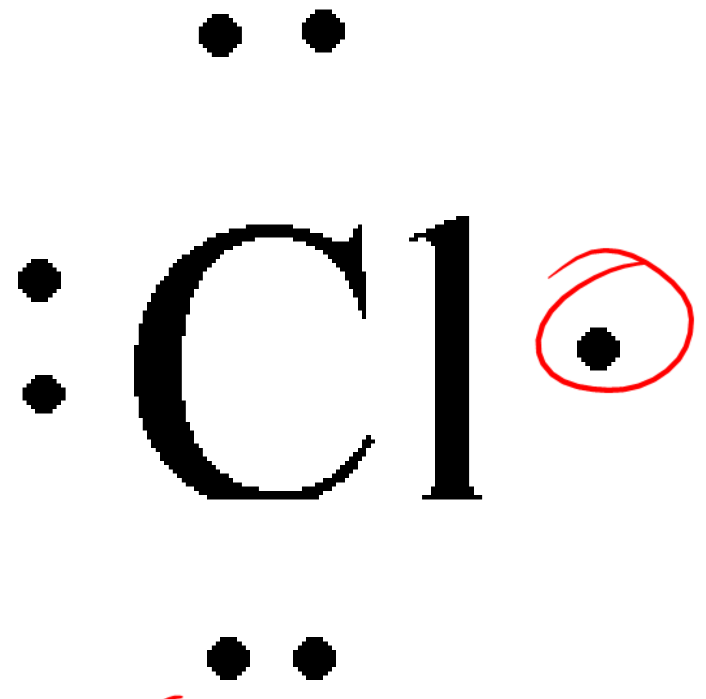
Lewis Dot Diagrams

only cares about valence e^-

1. Write the element symbol

Ar

2. Find the number of valence electrons by looking at the group number. For Groups 13-18, subtract 10 to obtain the valence electrons
3. Place dots representing the valence electrons on the four sides of the element symbol singly before pairing up



Cl

Checkpoint

Draw the Lewis Dot Diagram of:

a) Silicon

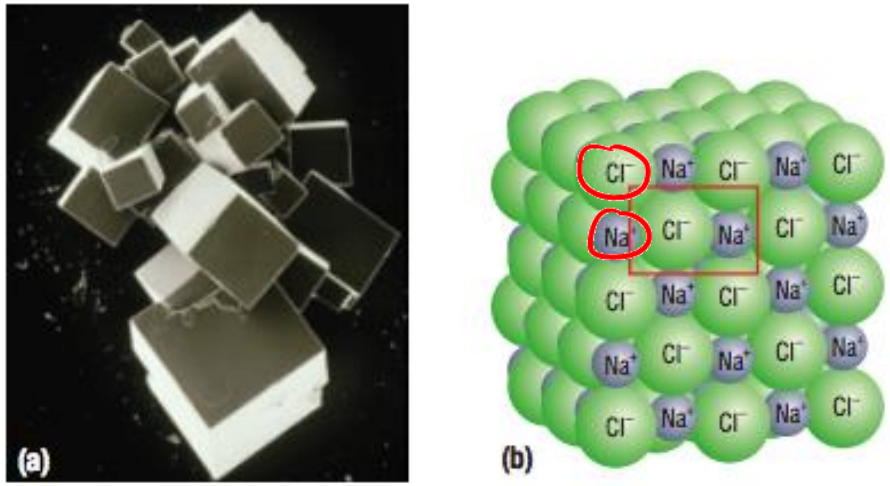
b) Ca

c) Mg^{2+}

d) O^{2-}

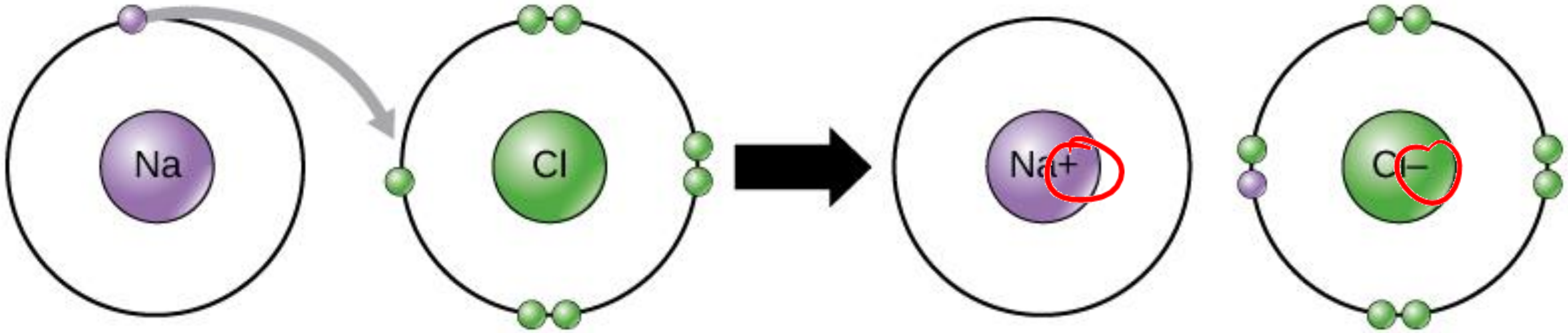
Ionic Compounds

- **Ionic Compounds** are formed by the electrostatic interaction of a cation and an anion; metal and a nonmetal



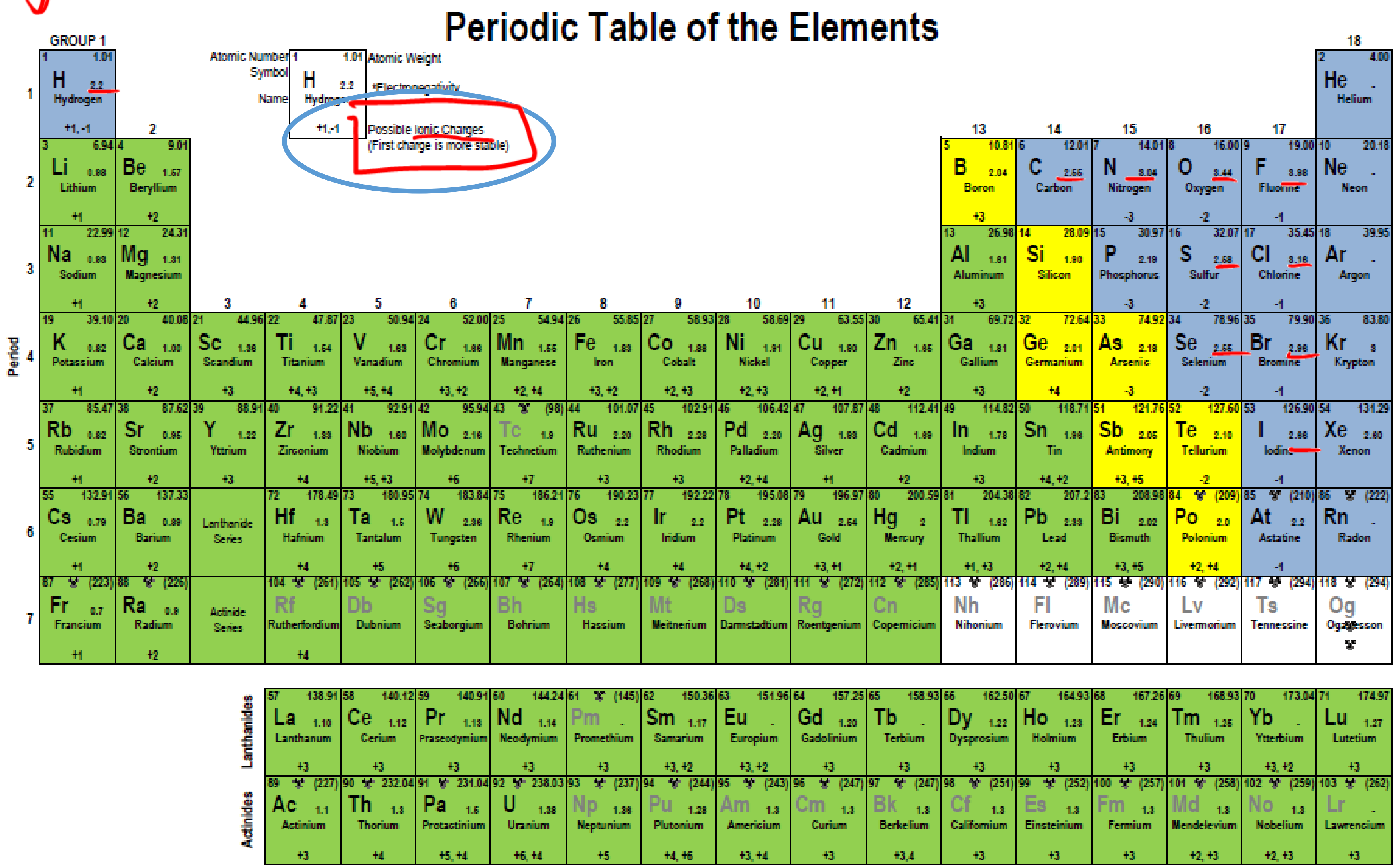
+ → ← −

NaCl



NaCl

exception H^+ usually



Metals tend to lose valence electrons

Non-metals tends to gain valence electrons

Physical Properties of Ionic Compounds

- High melting points and boiling points
- Low volatility/Non-volatile
- Generally soluble in polar solvents
- Do not conduct electricity in the solid state
- Conduct electricity when molten or when dissolved in water
- Generally brittle

Ionic compound	Charge on metal ion	Melting point
Na_2O	1+	1132 °C
MgO	2+	2800 °C

Writing Chemical Formulas

$\boxed{\overset{4+}{Mn}\ \overset{2-}{O_2}}$ $-4 = 0$

MnO_2

$\cancel{Mn_2O_4}$

Crisscross Method

1. Write the element symbols of the metal and then the nonmetal
2. Find the ionic charge of each element on the periodic table and write the ionic charge on the top-right corner
3. Criss-cross the ionic charge
4. Simplify the subscripts to the lowest terms and remove the charges

Ex: Calcium chloride

Ca Cl

Ca^{2+} Cl^{-}

Ca^{2+} Cl^{-}

Ca_1Cl_2

$CaCl_2$ no charge

Checkpoint

Draw the Lewis Dot Diagram for the formation of the following:

a) Magnesium bromide

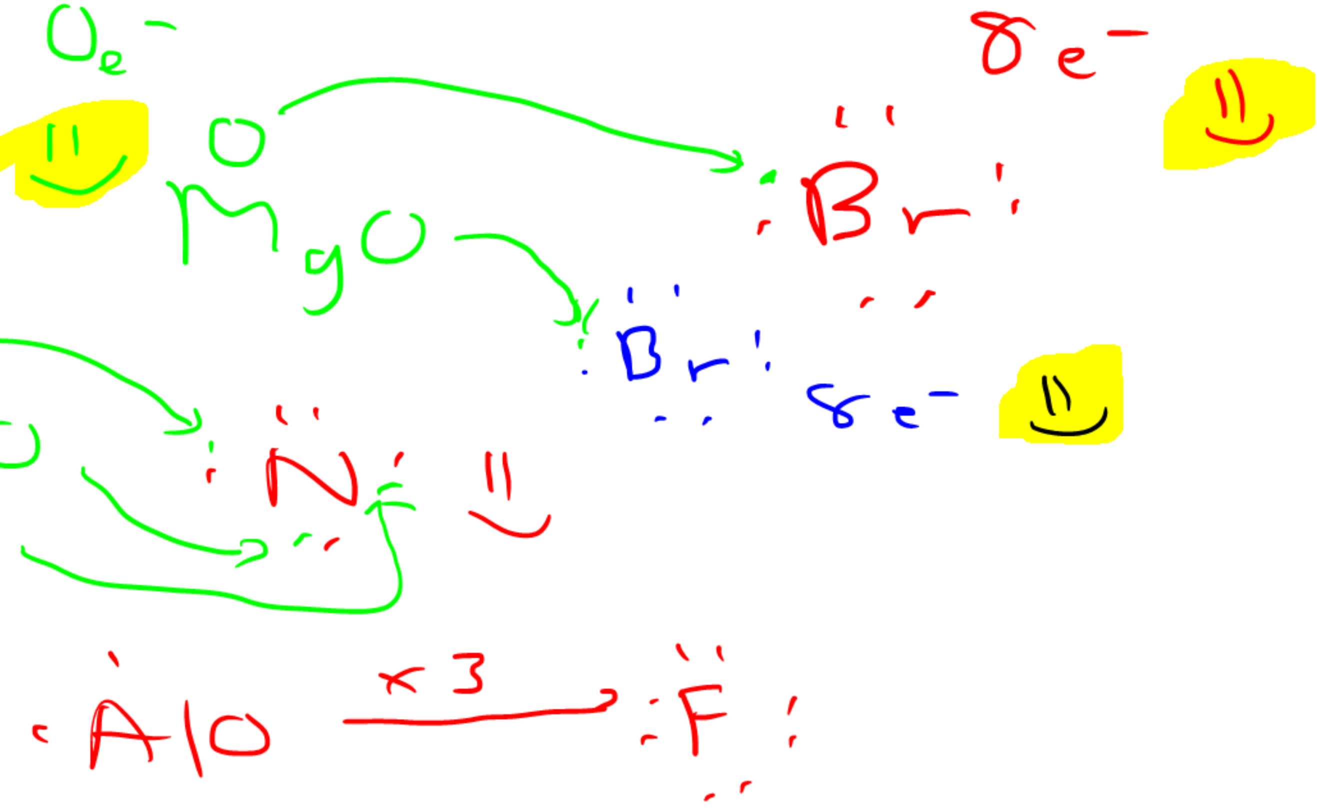
$MgBr_2$

b) Boron nitride

BN

c) Aluminum fluoride

AlF_3



Checkpoint

Write the chemical formula for the following:

a) Magnesium bromide $\boxed{Mg^{2+}\ Br^{-}_{2}}$ ✗ $MgBr_2$

b) Boron nitride $B^{3+}\ N^{3-}$ $B_3N_3 \rightarrow BN$

c) Aluminum fluoride Al^{3+} F^{-}_{3} AlF_3

safer

$\boxed{B^{3+}}$ B^{+3}

Writing Chemical Names

1. Un-crisscross the charges and consider if the compound was simplified. check periodic table
2. Write the name of the metal and check its ionic charge on the periodic table
3. Write the name of the nonmetal and check its ionic charge on the periodic table. For anions, change the ending to $\boxed{\textit{–ide}}$

Ex: BeO

$\boxed{Be^{1+}\ O^{1-}}$

this was simplified b/c O can't be -1

$Be^{2+}\ O^{2-}$

Beryllium ox~~ygen~~

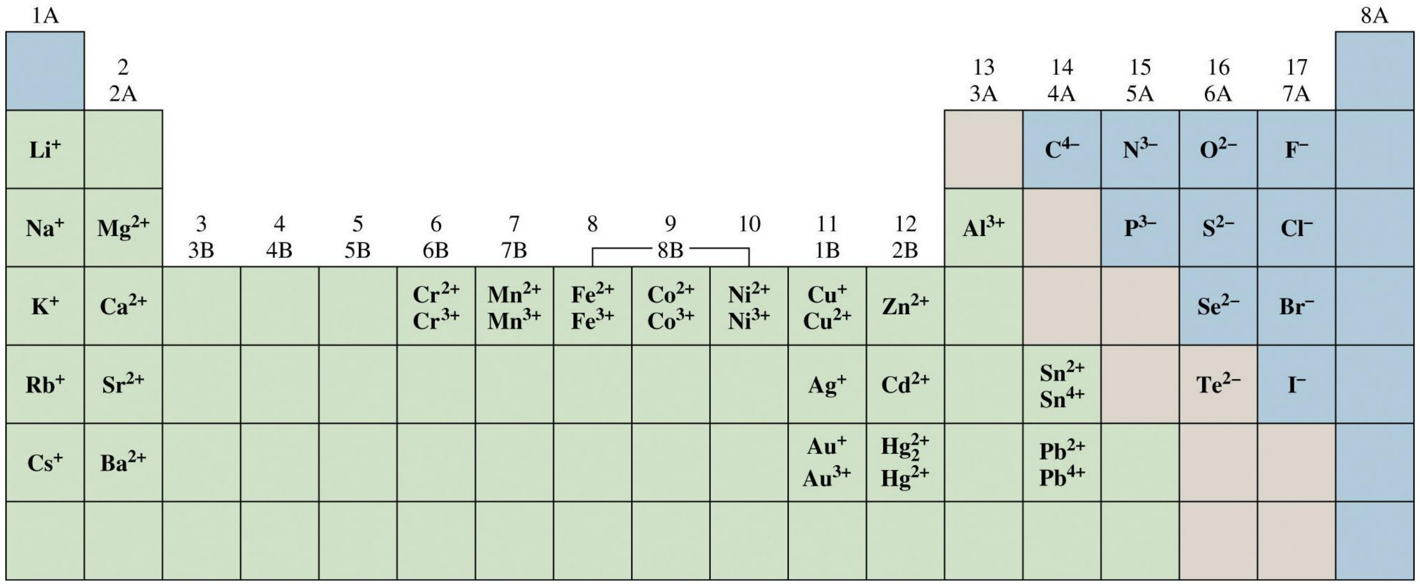
Beryllium oxide

I, II, III, IV, V, VI, VII, VIII, IX, X

1 2 3 4 5 6 7 8 9 10

Multivalent Elements

- Some transition metals are multivalent (more than one ionic charge)
- Use Roman numerals to distinguish between the ionic charges



Ion	Chemical Name
Fe^{2+}	Iron(II)
Fe^{3+}	Iron(III)
Cu^{+}	Copper(I)
Cu^{2+}	Copper(II)

Checkpoint

Write the chemical formula for the following:

$febr_3$ ✗

$FeBr_3$ ✓

a) Copper(II) chloride $CuCl_2$

b) Iron(III) bromide $Fe^{3+}\ Br^{-}$ $FeBr_3$

c) Nickel(II) oxide $Ni^{2+}\ O^{2-} = 0$ NiO

STOP Checkpoint STOP

Write the chemical name for the following:

a) Fe_2O_3 Fe^{3+} $\boxed{O^{2-}}$ Iron (III) oxide

b) Cu_2S Cu^{+} $\boxed{S^{2-}}$ Copper (I) sulfide ✓
sulphide ✓

c) Fe_1N_1 Fe^{1+} $\boxed{N^{1-}}$ ×3 Iron (III) nitride
Fe^{3+} N^{3-}

$Na_2(CO_3)$
1 2 3
> 2 assume polyatomic

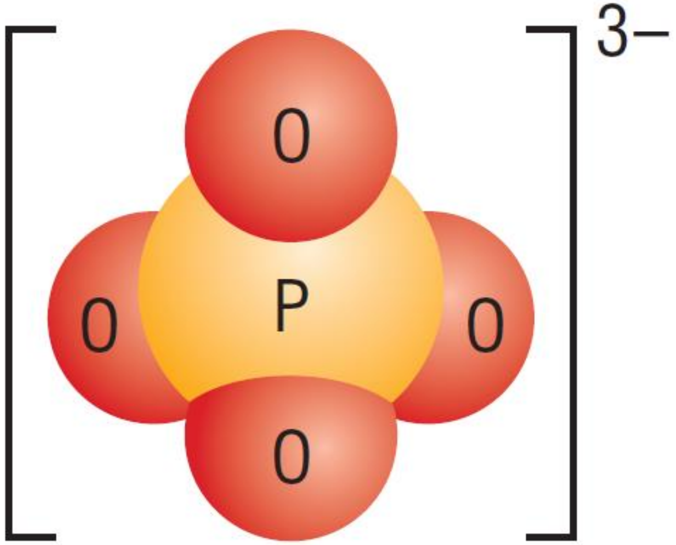
Polyatomic Ions

- **Polyatomic ion** is made of more than one atom and acts as a single unit

Ex: PO_4^{3-} phosphate ion

OH^- hydroxide ion

CO_3^{2-} carbonate ion



TIP: It may be helpful to keep the elements within the parentheses and the charge outside the parentheses. Ex: $(PO_4)^{3-}$ or $(OH)^-$

Writing Chemical Formulas with Polyatomic Ions

same as solving for NaCl

1. Write the metal with its ionic charge
2. Write the polyatomic ion with its charge outside the parentheses
3. Crisscross the ionic charges
4. Simplify the subscripts
   - Do not simplify the elements within the parentheses

Ex: Iron(III) nitrate

$Fe^{3+} (NO_3)^-$

$Fe^{3+} (NO_3)^-$

$\mathbf{Fe(NO_3)_3}$

Naming Chemical Names with Polyatomic Ions

1. Place parentheses around the polyatomic ion
2. Un-crisscross the charges
3. Write the name of the metal and check its ionic charge on the periodic table
4. Write the name of the polyatomic ion and check its ionic charge

from pg 2 of periodic table

Ex: Na_2CO_3

$Na_2(CO_3)$

$Na_2(CO_3)$

$Na^{1+}\ (CO_3)^{2-}$

Sodium carbonate

Checkpoint

doesn't end in -ide = polyatomic

Write the chemical formula for the following:

a) Calcium sulfate Ca^{2+} $(SO_4)^{2-}$ $CaSO_4$

b) Calcium chlorate Ca^{2+} $(ClO_3)^{-}_{2}$ $Ca(ClO_3)_2$

c) Iron(III) phosphate Fe^{3+} $(PO_4)^{3-}$ $FePO_4$

Checkpoint

Write the chemical name for the following:

a) KNO_3 K^+ $(NO_3)^-$ Potassium nitrate

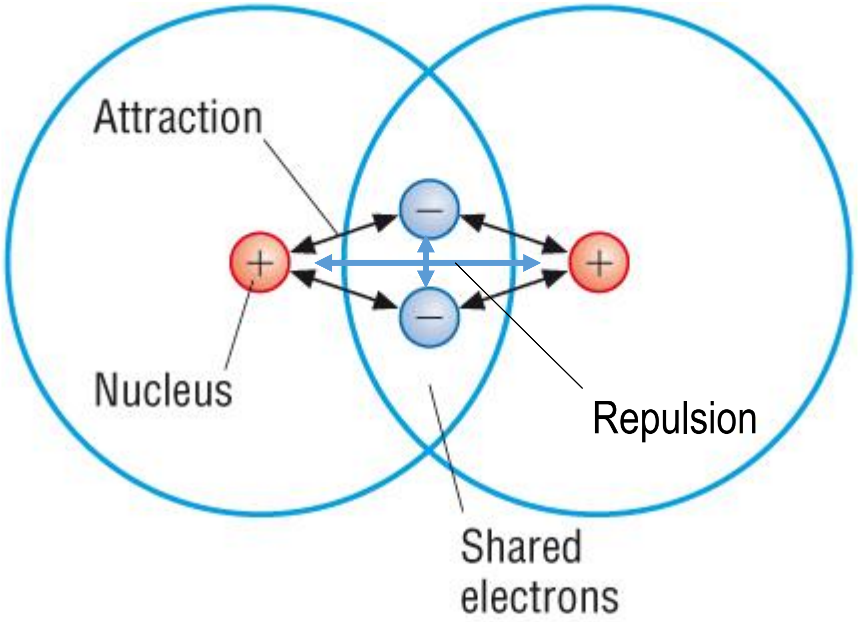
b) $(NH_4)_3PO_4$ Ammonium phosphate

c) $PbCO_3$ Lead (II) Carbonate

Pb^{2+} $(CO_3)^{2-}$

Covalent Bonding

- **Covalent bonding** is the stable balance of attractive forces and repulsive forces between atoms
  - Formed between nonmetals



a) Nitrous oxide (N_2O) is commonly known as laughing gas

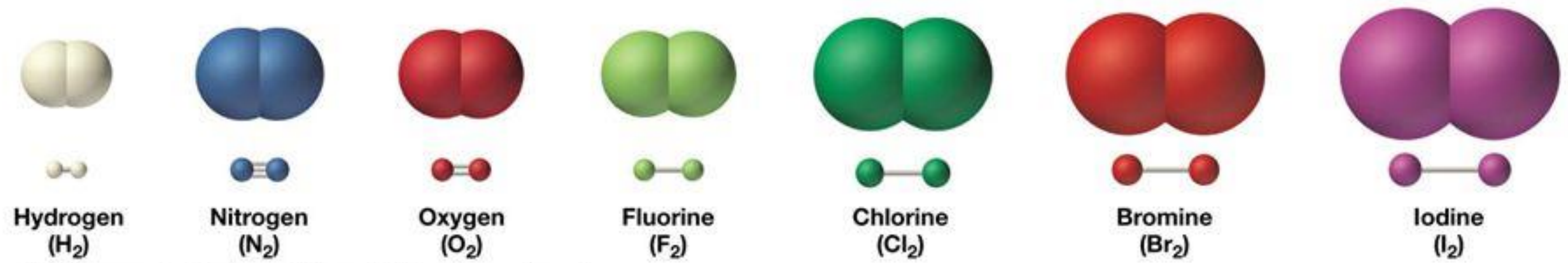
b) Nitrogen dioxide (NO_2) is the reddish-brown gas in smog

Physical Properties of Covalent Compounds

- Lower melting points and boiling points than ionic compounds
- Some are volatile (i.e. F_2, Cl_2, Br_2)
- Low solubility in water
- Do not usually conduct electricity as a solid, liquid, or in the aqueous state

Diatomic Molecules

- **Diatomic molecules** are molecules that consist of two atoms joined with covalent bonds
- Remember: **H_2 O_2 F_2 Br_2 I_2 N_2 Cl_2** ***Also P_4 and S_8**



(a) Elements that exist as diatomic molecules

Na

sodium oxide

Naming Covalent Molecules

1. Write the name of the elements
2. Change the ending of the second element to *-ide*
3. Add prefixes to represent the number of atoms
   - If mono- is the first prefix, it is understood and not written
   - If the element starts with a vowel, drop the (a) on the prefix

Ex: CO_2

carbon ox~~ygen~~

carbon oxide

carbon dioxide

Number	Prefix
1	Mon(o)-
2	Di-
3	Tri-
4	Tetr(a)-
5	Pent(a)-
6	Hex(a)-
7	Hept(a)-
8	Oct(a)-
9	Non(a)-
10	Dec(a)-

Checkpoint

Write the chemical names for the following:

a) CO carbon monoxide

b) PF_5

c) N_2O

Writing Molecular Formulas

1. Write the element symbol
2. Refer to the prefix and add subscripts to the element symbol
3. Do not simplify for covalent molecules

Ex: sulfur dioxide

S O

S_1 O_2

$\mathbf{SO_2}$

STOP Checkpoint STOP

Write the chemical formulas for the following:

a) Sulfur tetroxide SO_4

b) Dihydrogen dioxide H_2O_2

c) Carbon disulfide

Summary: Nomenclature

- **Naming Compounds**
  - **Ionic** (Metal + Nonmetal)
    - 1 ionic charge = metal + nonmetal-ide
    - >1 ionic charge = metal(IV) nonmetal-ide
    - Polyatomic = metal + polyatomic name*
  - **Covalent** (Nonmetal + Nonmetal)
    - = pre-nonmetal + pre-nonmetal

*If NH_4^+, then $(NH_4)^+$ + nonmetal-ide

What I Learned Today:

- ❑ Lewis Dot Diagrams
- ❑ Physical Properties of Ionic Compounds
- ❑ Naming and Writing Ionic Compounds with:
    - Single charge
    - Multivalent charges
    - Polyatomic ions
- ❑ Naming and Writing Covalent Molecules

Due next class: Class 1 Homework